Azra Holland & Emily Satterfield

LIS 5643 - Final Project

December 8, 2025

Predicting Diabetes Status Using Health and Lifestyle Indicators

1. Problem Statement

The purpose of this project is to develop a predictive model that can classify an individual's diabetes status (either no diabetes, prediabetes, or diabetes) based on several health and lifestyle behaviors. Diabetes is a prevalent public health issue, and identifying early indicators can support timely intervention and prevention. Because the target variable (Diabetes_012) consists of three categorical outcomes, this analysis uses a multiclass classification approach.

2. Dataset Description and Relevance

The dataset used in this analysis is the Diabetes 012 Health Indicators BRFSS 2015 dataset from the CDC, which includes self-reported survey responses about health behaviors, medical history, and demographic factors. The dataset contains variables such as:

- BMI (Body Mass Index)
- HighBP (High blood pressure diagnosis)
- HighChol (High cholesterol diagnosis)
- PhysActivity (Engagement in physical activity)
- Smoker (Smoking history)

- Age
- GenHlth (Self-rated general health)
- Diabetes_012 (target variable)

These variables were chosen because they are widely recognized risk factors for diabetes.

3. **Methodologies and Techniques Used**

Data exploration and understanding

The analysis began by loading the dataset and examining its structure using summary statistics. To better understand data distribution and relationships, several visualizations were generated: a histogram of BMI to examine its distribution, a boxplot of BMI vs. diabetes status to highlight group differences, bar charts showing high blood pressure and physical activity across diabetes categories.

Data preprocessing

Imputation was unnecessary because the dataset contained no missing values. Because the target variable represents classification groups, it was converted to a factor. BMI was scaled using min-max scaling to normalize differences in range between BMI and other binary predictors. The dataset was split into 70% training and 30% testing subsets using stratified sampling to ensure balanced representation of diabetes categories in both sets.

Data mining and exploration

- **Correlation analysis**
    - To identify linear relationships between predictors and diabetes status, simple correlation coefficients were calculated between diabetes status (that was converted to numeric) and several numeric predictors, including BMI, age, high blood pressure, and high cholesterol. These values were displayed in a small table and visualized as a bar plot. Positive correlations suggested predictors tend to increase with diabetes severity.
- **Association rule mining**
    - The Apriori algorithm was applied to several predictors, including high blood pressure, high cholesterol, smoking status, physical activity, general health, and diabetes status. Because the dataset contains mostly binary variables with limited item combinations, the support threshold had to be set very low (0.0005) to extract meaningful rules. After lowering the thresholds, the algorithm successfully generated association rules. Several rules revealed expected patterns, such as individuals with diabetes being more likely to also report high blood pressure or high cholesterol. Rules involving poor general health also appeared frequently, reinforcing known relationships between chronic health conditions and health perceptions.

Key Predictors Identified

Based on correlation analysis, domain knowledge, and the structure of the decision tree model, the strongest predictors for diabetes in this dataset include high blood pressure, high cholesterol,

BMI, general health rating, age, and physical activity. These findings are consistent with known public health research on diabetes risk factors.

Model Selection and Application

Because the target variable includes three categorical classes, a multiclass classification model is required. A decision tree classification model was selected because it handles mixed categorical and numeric variables, and it produces clear, interpretable decision paths. The model was trained on the training dataset using rpart(). To improve generalization, the model's complexity parameter table (cp table) was reviewed, and the tree was pruned using the cp value associated with the lowest cross-validated error. Both the initial and pruned trees were visualized. Because the pruned tree and the initial tree were identical, this indicates that the initial tree was already as simple as possible based on the dataset.

Model Evaluation and Diagnostics

Model performance was evaluated using predictions and accuracy. Accuracy was calculated manually for both the training and test sets. Training accuracy indicates how well the model learned from the training data**,** and testing data indicates how well the model generalizes to new, unseen data.

Comparing these two values provides insight into overfitting. In this analysis, the training and testing accuracies were similar, suggesting that the pruned decision tree generalizes reasonably well and is not severely overfitting.

4. Results, Insights, and Implications

The analysis produced several meaningful insights:

- Higher BMI, high blood pressure, and high cholesterol were associated with higher diabetes categories, reinforcing their importance as risk factors.
- Physical inactivity and poor general health ratings were also connected to diabetes severity.
- The decision tree model demonstrated reasonable predictive accuracy and provided interpretable decision paths.
- The inability of the Apriori algorithm to generate rules highlights the structural limitations of the dataset rather than methodological issues.

These insights support existing public health knowledge about diabetes risk factors .

5. Recommendations and Conclusions

Based on this project's results, the following recommendations are offered:

1. Public health initiatives could focus on improving physical activity, weight management, and blood pressure control to reduce diabetes risk.

2. Predictive modeling could be used in health clinics or community health programs to identify individuals at risk based on easily collected variables.

Conclusion:

The resulting model gotten from doing this project provides meaningful insights into diabetes risk indicators and demonstrates how data analytics can support decision-making in health contexts.